INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT I

Electric Vehicle Propulsion and Energy Sources

I. Introduction to Electric Vehicles

An **Electric Vehicle (EV)** is a vehicle that uses one or more **electric motors** for propulsion, with energy stored in a battery (or fuel cell / ultracapacitor) instead of burning fuel in an engine only.

Classification of EVs

- **BEV** (Battery EV) - only battery + motor, zero tail-pipe emission. Ex: Nissan Leaf, Tesla.
- **HEV** (Hybrid EV) - IC engine + electric motor, battery charged by engine / regenerative braking.
- **PHEV** (Plug-in Hybrid) - HEV with larger battery that can be charged from the grid.
- **FCEV** (Fuel-cell EV) - hydrogen fuel cell produces electricity for the motor. Ex: Toyota Mirai.

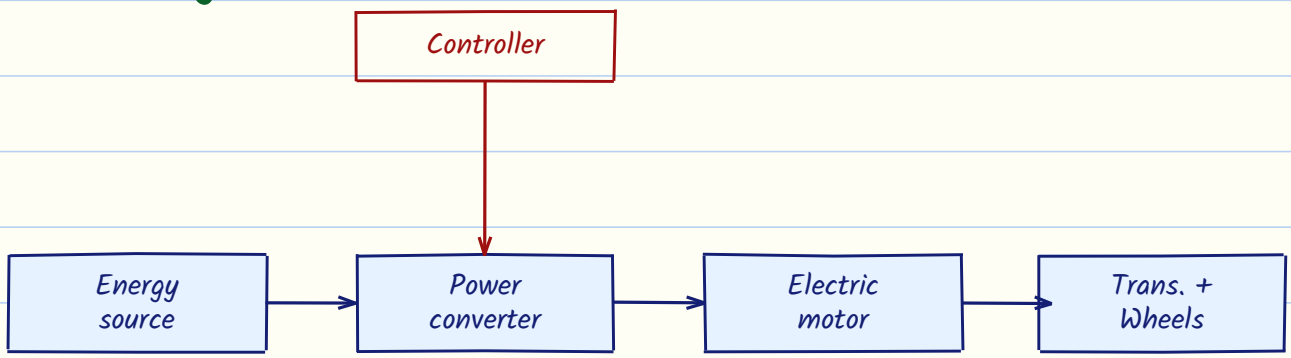
Advantages

- High efficiency (motor 85-95% vs engine 25-35%), no local pollution, low noise.
- Regenerative braking recovers kinetic energy; instant torque; low running and maintenance cost.

Limitations

- Low specific energy of batteries -> limited range; long charging time; high initial cost; charging infrastructure needed.

Block diagram of an electric drive train



Basic EV drive train (energy flows left to right; regen flows back)

The **energy source** (battery) supplies power, the **power converter** (inverter / DC-DC) controls voltage and current, the **motor** converts electrical to mechanical power, and the **transmission** (single-speed gear) drives the wheels. The **controller** takes accelerator / brake commands and drives the converter.

2. Vehicle Mechanics - Kinetics and Dynamics

Kinetics deals with the forces causing motion; **dynamics** deals with how the vehicle moves under those forces (Newton's second law). For a vehicle moving along a road, the tractive force from the wheels must overcome all the resistive forces.

Newton's second law applied to the vehicle

$$m\,(dv/dt) = \Sigma F_t - \Sigma F_r$$

where ΣF_t = total tractive force at the driven wheels and ΣF_r = sum of the resistive forces (rolling, aerodynamic, grade). Here m = vehicle mass (kg), v = speed (m/s).

Tractive force at the wheel

$$F_t = T_m\, G\, \eta_t / r_w$$

- T_m = motor torque (N-m), G = overall gear ratio, η_t = transmission efficiency, r_w = wheel radius (m).
- Wheel speed $\omega_w = v / r_w$ and motor speed $\omega_m = G\,\omega_w = G\,v / r_w$.
- Tractive force is limited by adhesion: $F_{t,max} = \mu_a N$, where N = normal load on driven wheels and μ_a = tyre-road adhesion coefficient (about 0.8 dry road, 0.2-0.3 wet, 0.1 ice).

3. Roadway Fundamentals

The road affects the vehicle through its **surface (rolling resistance)** and **gradient (slope)**.

(a) Rolling resistance F_{rr}

Caused by hysteresis (deformation) of the tyre and the road surface. It is almost independent of speed at low speeds.

$$F_{rr} = \mu_{rr}\, m\, g \cos\alpha$$

- μ_{rr} = rolling resistance coefficient: 0.008-0.015 for car tyres on concrete / asphalt, about 0.03-0.04 on gravel, 0.2-0.3 on sand.
- Depends on tyre pressure, tyre material and temperature;

low-rolling-resistance tyres are used in EVs.

(b) Aerodynamic drag F_{ad}

$$F_{ad} = \tfrac{1}{2}\,\rho\, C_D\, A_f\, (v + v_w)^2$$

- ρ = air density (1.2 kg/m³), C_D = drag coefficient (0.25–0.4 for cars), A_f = frontal area (m²), v_w = head-wind speed.
- Drag grows with the **square of speed**, so drag power grows with the cube of speed – dominant on highways.

(c) Grading (hill climbing) resistance F_{hc}

$$F_{hc} = m\, g\, \sin\alpha$$

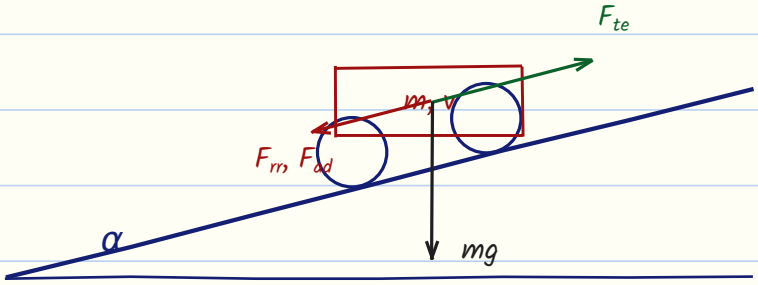
- α = road slope angle. Grade (%) = 100 tanα. For small angles sin$\alpha \approx \tan\alpha$, so 5% grade means sin$\alpha \approx 0.05$.
- Positive on uphill (resists), negative on downhill (assists motion).

(d) Acceleration force F_{la} and rotational inertia

$$F_{la} = m\, a \;;\; F_{\omega a} = (J/r_w^2)\, a$$

The rotating parts (motor rotor, gears, wheels) also need to be accelerated. This is included by a **rotational inertia factor** $\delta = 1 + (J_{eq}/(m\, r_w^2))$ giving $F_{la} = \delta\, m\, a$, where $\delta \approx 1.05$–1.1.

Resistive forces on a slope:
rolling $F_{rr} = \mu_{rr}\, m\, g \cos\alpha$
aerodynamic $F_{ad} = \tfrac{1}{2}\, \rho\, C_D\, A_f\, v^2$
hill climb $F_{hc} = m\, g \sin\alpha$
acceleration $F_{la} = m\, a$



4. Propulsion System Design – Force-Velocity Characteristics

Total tractive effort

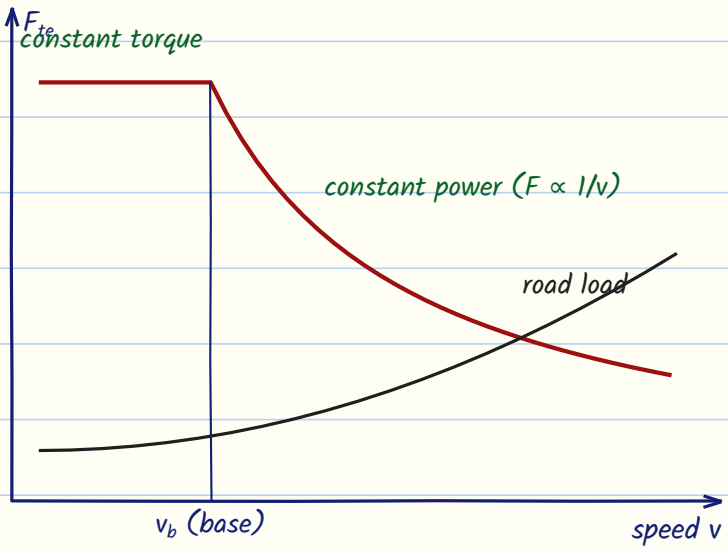
$$F_{te} = F_{rr} + F_{ad} + F_{hc} + F_{la} + F_{\omega a}$$

Propulsion design means choosing the motor, gear ratio and battery so that the EV meets the specified **top speed, acceleration (0–100 km/h) and gradeability**.

Ideal force-velocity characteristic

An ideal traction motor gives **constant torque** from zero speed up to the base speed v_b, and **constant power** beyond it. In the constant-torque region the force is constant; in the constant-power region $F = P / v$, so force falls hyperbolically with speed.

$$P_{te} = F_{te}\, v$$



Intersection of the two curves = maximum speed of the EV.

Gap between curves = force left for acceleration and gradeability.

- A multi-speed gearbox is not required for electric motors with wide constant-power range - single speed reduction ($G = 8\text{-}10$) is enough.
- The constant-power region ratio $x = \omega_{max}/\omega_{base}$ is typically 2-4 for induction/PM machines. Larger x means smaller motor torque rating for the same power.
- The gear ratio is chosen from top speed: $G = \omega_{m,max}\, r_w / v_{max}$.

5. Calculation of Tractive Power and Energy

Tractive power and energy formulas

$$P_{te}(t) = F_{te}(t) \cdot v(t)$$

$$E = \int P_{te}\, dt \text{ (traction energy at wheels)}$$

- Battery power $P_{batt} = P_{te} / (\eta_t\, \eta_m\, \eta_c)$ during motoring; during regeneration the efficiencies multiply instead ($P_{batt} = P_{te}\, \eta\, \eta\, \eta$).
- Energy consumption is expressed in Wh/km = E / distance. Range = usable battery energy / Wh per km.

Worked example 1 – cruising on flat road

Data: $m = 1500$ kg, $\mu_{rr} = 0.012$, $\rho = 1.2$, $C_D = 0.3$, $A_f = 2.2$ m², $v = 100$ km/h $= 27.78$ m/s, $\alpha = 0$.

- $F_{rr} = 0.012 \times 1500 \times 9.81 =$ **176.6 N**
- $F_{ad} = 0.5 \times 1.2 \times 0.3 \times 2.2 \times (27.78)^2 =$ **305.5 N**
- $F_{te} = 176.6 + 305.5 = 482$ N -> $P_{te} = 482 \times 27.78 =$ **13.4 kW**
- Energy for 10 km (time 0.1 h) $= 13.4 \times 0.1 =$ **1.34 kWh**, i.e. about 134 Wh/km at wheels.

Worked example 2 – climbing 5% grade at 60 km/h

- $v = 16.67$ m/s; $F_{rr} = 176.6$ N; $F_{ad} = 0.396 \times (16.67)^2 = 110$ N; $F_{hc} = 1500 \times 9.81 \times 0.05 = 735.8$ N
- $F_{te} = 1022$ N -> $P_{te} = 1022 \times 16.67 =$ **17.0 kW**

Worked example 3 – acceleration

- 0 to 100 km/h in 10 s on flat road -> $a = 27.78/10 = 2.78$ m/s²; $F_{la} = 1500 \times 2.78 = 4167$ N
- Peak power at the end of acceleration (ignoring δ) $= (4167 + 482) \times 27.78 \approx$ **129 kW**. So a motor of about 130 kW peak is needed.

> Remember: rolling resistance dominates at low speed, aerodynamic drag at high speed, and gradient/acceleration force decides the peak power rating of the motor.

6. Electric Vehicle Power Source - Battery Parameters

Terminology

- **Battery capacity (Q)** - total charge the battery can deliver, in ampere-hours (Ah), measured at a stated C-rate, temperature and cut-off voltage. Energy capacity (Wh) = $Q \times V_{nominal}$.
- **C-rate** - discharge current relative to capacity: I = (C-rate) × Q. 1C for a 50 Ah cell = 50 A; 0.5C = 25 A.
- **State of Charge (SOC)** - remaining capacity as % of rated capacity.

$$SOC = (Q_{remaining} / Q_{rated}) \times 100\ \%$$
$$DOD = 100\ \% - SOC \text{ (Depth of Discharge)}$$

- **State of Discharge** - the same quantity seen from the other side: amount of charge already removed (DOD). SOC = 100% means fully charged, DOD = 100% means empty.
- **Specific energy** (Wh/kg) - energy stored per unit mass; decides the **range**. **Energy density** is per unit volume (Wh/L).
- **Specific power** (W/kg) - maximum power delivered per unit mass; decides **acceleration** and gradeability.
- **Terminal voltage, internal resistance, cycle life, self-discharge, efficiency** (Ah and Wh efficiency).

Peukert's relation (effect of discharge rate on capacity)

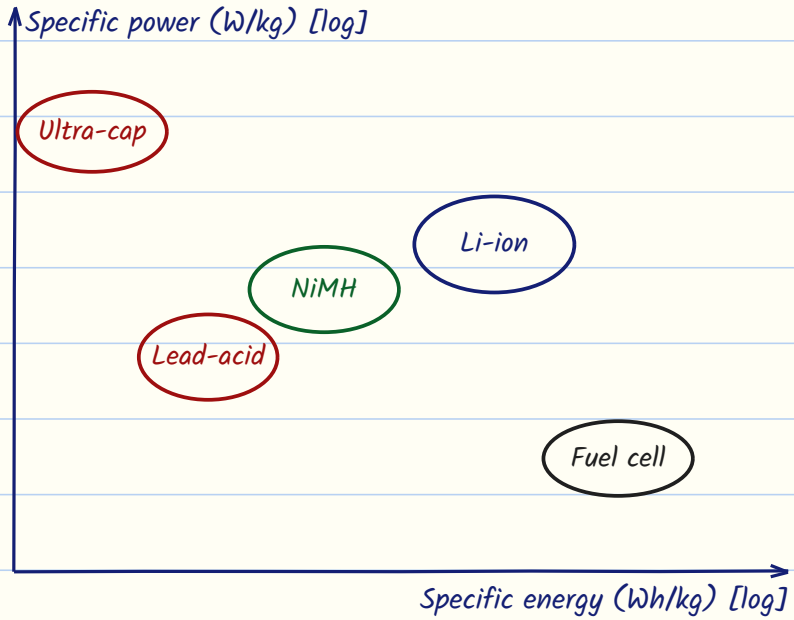
$$t = H\ (\ C\ /\ (I\ H)\ )^{k}$$

H = rated discharge time (h), C = rated capacity at H hours, I = actual current, k = Peukert constant (1.05-1.3 for lead-acid, about

1.0-1.1 for Li-ion). Higher current -> less available capacity.

Ragone plot

A **Ragone plot** is a log-log graph of **specific power (y-axis) against specific energy (x-axis)** for energy storage devices. It lets us compare sources: a device cannot give both very high energy and very high power. EVs need both, so batteries are often combined with ultracapacitors.



Ragone plot:
shows power vs energy trade-off of each source. Top-left: high power, low energy (ultracap). Bottom-right: high energy, low power (fuel cell).

7. Battery Modeling

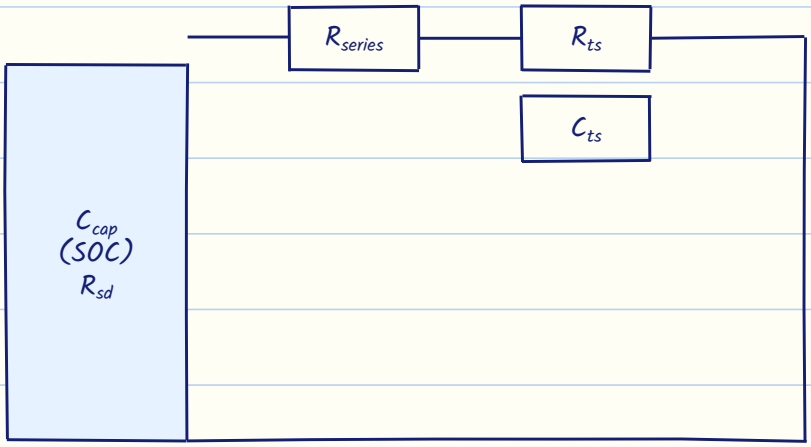
A battery model predicts terminal voltage, SOC and run time for a given load current. Used in simulation, BMS and EV range estimation.

(a) Run-time battery model

An electrical-circuit (equivalent circuit) model. It predicts both the **transient response** and the **run time** (how long the battery lasts) for a given load. Components: a **capacitor C_{cap}** whose

voltage represents SOC, a self-discharge resistor R_{sd}, a voltage source $V_{oc}(SOC)$, series resistance R_{series} and RC networks for transients. Parameters are fitted from discharge tests and depend on SOC, temperature and cycle number.

Run-time battery model (Thevenin / Chen-Rincon-Mora type)



$V_{oc}(SOC)$: open circuit
R_{series} : ohmic drop
R||C : transient (short term)
C_{cap} : stored charge = SOC
R_{sd} : self-discharge

- Run time = time taken for the SOC voltage on C_{cap} to fall to the cut-off value under the load current.
- Simple, fast, easy to put in vehicle simulations (MATLAB / Simulink, ADVISOR).

(b) First-principle (electrochemical) model

Based on the physics and chemistry inside the cell: electrode reactions, ion diffusion in electrolyte and solid particles and charge conservation (porous electrode theory, Doyle-Fuller-Newman model).

- Open-circuit voltage from the **Nernst equation**: $E = E_0 - (R\,T / n\,F) \ln Q_r$.
- Reaction kinetics - **Butler-Volmer equation**: $i = i_0[\exp(\alpha_a F \eta$

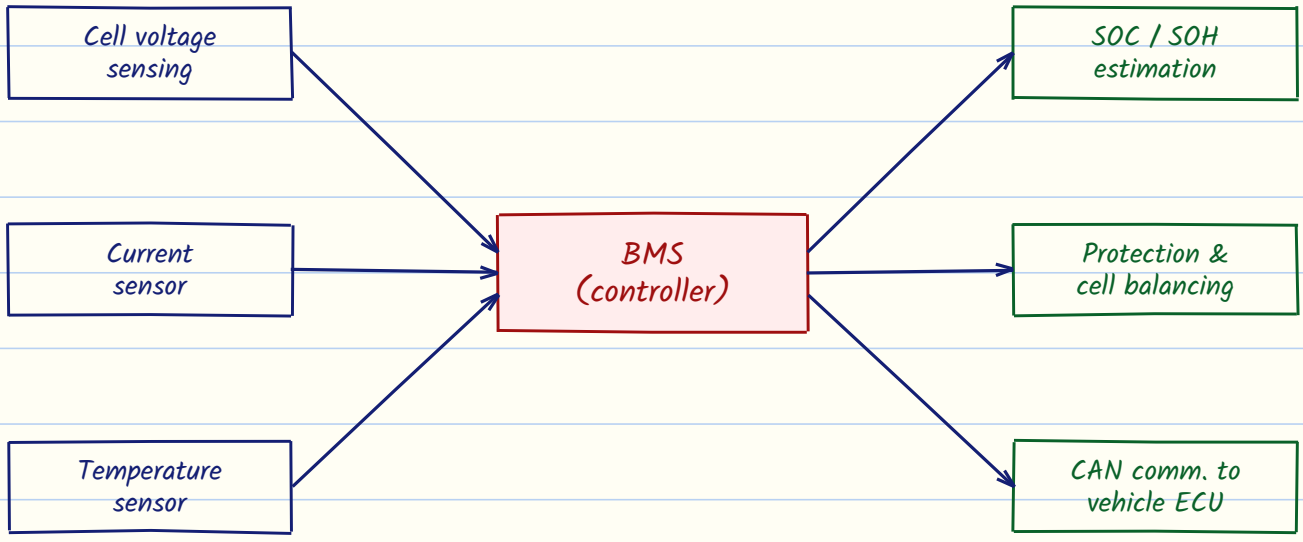
$/ R T) - \exp(-\alpha_c F \eta / R T)]$, where η = overpotential.

- Diffusion of lithium ions in the solid and electrolyte - **Fick's law**: $\partial c/\partial t = D \nabla^2 c$.
- Very accurate and gives internal states (concentration, temperature) but needs many parameters and heavy computation - used for cell design, not in real-time BMS.

Feature	Run-time (circuit) model	First-principle model
Basis	Electrical analogy, curve fitting	Electrochemistry, physics
Accuracy	Moderate	High
Computation	Light, real-time	Heavy
Use	BMS, vehicle simulation	Cell design / research

8. Battery Management System (BMS)

The **BMS** is the electronic system that monitors and protects the battery pack and keeps it in its safe operating area.



Functions of BMS

- Measures cell voltages, pack current, temperatures; protects against over-charge, over-discharge, over-current and over/under temperature.
- Estimates **SOC** (fuel gauge) and **SOH** (state of health), calculates remaining range and available power.
- Cell balancing, thermal management (fan / liquid cooling control), contactor control and fault diagnosis; communicates on CAN bus.

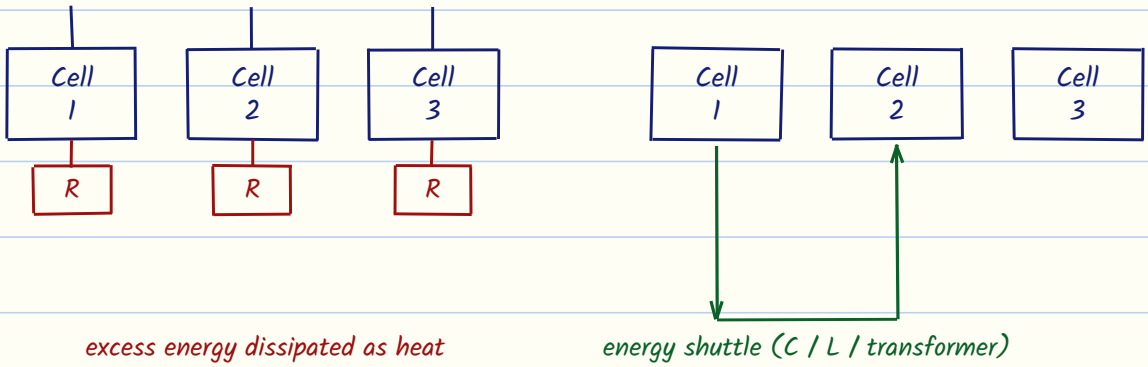
SOC measurement methods

- **Coulomb counting (ampere-hour integration)**: $SOC(t) = SOC(0) + (1/Q_n) \int \eta\, I\, dt$ (I positive on charge). Simple but drifts due to current-sensor error and unknown initial SOC.
- **Open-circuit voltage method**: SOC read from the V_{oc}-SOC curve after the battery rests. Accurate but needs rest time; flat curve for LiFePO4 / NiMH makes it hard.
- **Impedance / internal-resistance method** and (for lead-acid) specific gravity of electrolyte.
- **Kalman filter / observer-based method**: combines model and measured V, I to give an optimal SOC estimate that corrects drift - used in modern EVs.
- Practical BMS uses coulomb counting + OCV correction + Kalman filter together.

Battery cell balancing

Cells in a series string differ slightly in capacity, internal resistance and self-discharge, so some cells reach full / empty earlier. The weakest cell then limits the pack. **Cell balancing** equalises the SOC of all cells.

(a) Passive balancing (b) Active balancing



- **Passive balancing** - a bleed resistor across the higher-voltage cell burns the excess charge as heat. Cheap, simple but wasteful; used mostly during charging.
- **Active balancing** - transfers charge from strong cells to weak cells using capacitors (switched capacitor), inductors (buck-boost) or transformers. Efficient, faster, costlier and more complex.

9. Traction Batteries

(a) Nickel-Metal Hydride (NiMH) battery

- Positive electrode: nickel oxyhydroxide NiOOH; negative: hydrogen-absorbing metal alloy (MH); electrolyte: KOH. Nominal cell voltage **1.2 V**.

- Reaction: $NiOOH + MH <-> Ni(OH)_2 + M$ (charge <-> discharge). No cadmium, so environment-friendly.
- Specific energy 60-120 Wh/kg, specific power 200-1000 W/kg, life 500-1500 cycles, high self-discharge, good safety, wide temperature range.
- Used in Toyota Prius (first and second generation) and many hybrid cars.

(b) Lithium-Ion (Li-ion) battery

- Cathode: lithium metal oxide (LCO, NMC, NCA, LFP); anode: graphite; electrolyte: lithium salt ($LiPF_6$) in organic solvent; separator between. Nominal voltage 3.2-3.7 V.
- During charge Li^+ ions move from cathode to anode (intercalate in graphite); on discharge they return and electrons flow through the external circuit.
- Specific energy 100-265 Wh/kg, specific power 300-1500 W/kg, life 1000-3000 cycles, low self-discharge (2-5% per month), no memory effect, efficiency above 95%.
- Needs BMS: overcharge / over-temperature can cause thermal runaway.

(c) Lithium-Polymer (Li-poly) battery

- A Li-ion variant that uses a **gel / solid polymer electrolyte** instead of liquid; cells are thin flat pouch type - flexible shape and light weight.
- Specific energy 130-200 Wh/kg, lower risk of leakage, better

packaging, but lower conductivity at low temperature and higher cost; life 500-1000 cycles.

Comparison

Parameter	Lead-acid	NiMH	Li-ion	Li-poly
Cell voltage (V)	2.0	1.2	3.6	3.7
Specific energy (Wh/kg)	30-40	60-120	100-265	130-200
Specific power (W/kg)	~180	200-1000	300-1500	~250-2000
Cycle life	200-500	500-1500	1000-3000	500-1000
Self discharge	Low	High (~30%/mo)	Very low	Very low
Cost	Lowest	Medium	High	Highest

Important questions: (1) Derive the tractive effort equation. (2) Explain Ragone plot. (3) SOC estimation methods. (4) Cell balancing. (5) Compare NiMH, Li-ion and Li-poly batteries.

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UNIT II

Electric Vehicle Power Plant and Drives

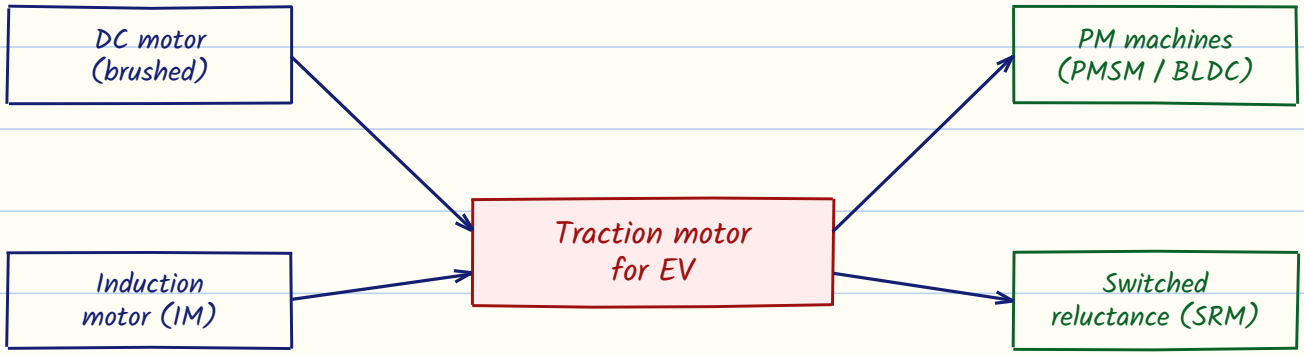
I. Introduction - Electric Vehicle Power Plants

The **power plant** of an EV is the electric **propulsion system**: battery -> power electronic converter -> electric motor -> transmission. Here the "engine" is the **electric machine plus its power converter and controller**.

Requirements of an EV traction motor

- High torque at low speed (starting, hill climbing) and high power at high speed - wide constant-power range (3-4 times base speed).
- High **efficiency** over a wide torque-speed range, especially at light loads (drive cycles).
- High power density and torque density (small size, low weight), good cooling.
- Regenerative braking capability, fast torque response, robustness, low noise, low cost, high reliability.

Types of electric machines used



Early EVs used brushed **DC motors** (simple control but brushes need maintenance). Modern EVs use brushless machines: **induction**, **permanent magnet (PM)** and **switched reluctance (SRM)** motors with inverter control.

2. Induction Machines (IM)

The most widely used machine in EVs (e.g. Tesla Model S early versions). It has a stator with three-phase winding and a **squirrel-cage rotor** (bars short-circuited by end rings).

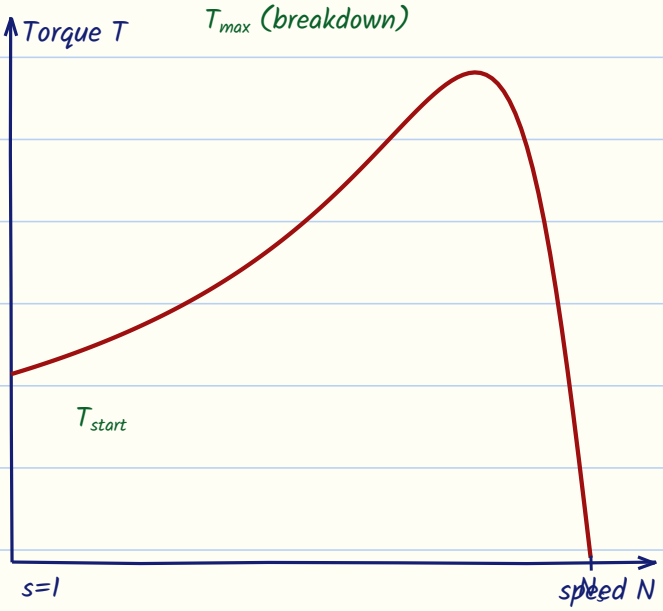
Working principle

- The stator three-phase currents create a **rotating magnetic field** at synchronous speed $N_s = 120\ f / p$ (rpm).
- The field cuts the rotor conductors and induces emf and current in them (transformer action). Torque is produced

by interaction of the rotor current with the air-gap flux.

- The rotor runs slightly below N_s. The difference is called **slip**: $s = (N_s - N) / N_s$. Torque exists only when $s \neq 0$.

$$T = (3\,p / 2\,\omega_s) \cdot V^2 (R_2/s) / [\,(R_1 + R_2/s)^2 + (X_1 + X_2)^2\,]$$



Slip $s = (N_s - N)/N_s$

Sync speed $N_s = 120\,f / p$

Stable region: between T_{max} and N_s

Speed control: V/f constant shifts the curve sideways.

Control of IM in EVs

- **Constant V/f control** (scalar) - keeps flux constant below base speed. Simple but poorer dynamic response.
- **Field-oriented (vector) control, FOC** - stator current split into flux-producing (i_d) and torque-producing (i_q) components controlled independently like a DC motor; **field weakening** above base speed gives constant-power region.
- **Direct torque control (DTC)** - directly controls torque and flux with hysteresis controllers; fast response.

Merits and demerits

- + Simple, rugged, low cost, no magnets, low maintenance, good field-weakening range, reliable at high speed.
- - Lower efficiency than PM motors (rotor copper loss), lower power factor, rotor heating, control more complex, lower power density.

3. Permanent Magnet (PM) Machines

Magnets (NdFeB, SmCo) in the rotor give excitation flux without field current, hence no rotor copper loss - **highest efficiency and power density** among EV motors. Used in Toyota Prius, Nissan Leaf, Chevrolet Volt.

Types

- **PM synchronous motor (PMSM)** - sinusoidal back-emf, sinusoidal stator current; smooth torque, controlled by FOC.
- **PM brushless DC motor (BLDC)** - trapezoidal back-emf, rectangular current in two phases at a time using rotor position (Hall) sensors; simple control, some torque ripple.

Rotor construction

- **Surface-mounted (SPM)** - magnets glued on rotor surface; low saliency, limited field weakening.
- **Interior (IPM)** - magnets buried inside rotor; gives

reluctance torque also: $T = (3p/2)[\psi_m i_q + (L_d - L_q) i_d i_q]$; wide field-weakening range; most popular in EVs.

Merits and demerits

- + High efficiency, high torque / power density, good power factor, compact.
- - High magnet cost, magnets can demagnetise at high temperature, limited constant-power range, back-emf at high speed (needs field weakening current), uncontrolled generation fault risk.

4. Switched Reluctance Machines (SRM)

A **doubly-salient** machine: both stator and rotor have teeth (salient poles). The stator has concentrated coil windings, the rotor has **no winding and no magnets** - just laminated steel. Common types 6/4 (3-phase) and 8/6 (4-phase).

- Torque is produced by the tendency of the rotor to align with the energised stator pole to **minimise reluctance** (maximise inductance).
- Torque is independent of the direction of current: $T = \frac{1}{2} i^2 (dL/d\theta)$. Phases are excited in sequence, synchronised with rotor position (shaft encoder needed).
- + Very rugged and cheap, high-speed capability, fault tolerant (phases independent), wide constant-power

range, no magnet cost.

- – **Torque ripple**, acoustic noise and vibration, needs special converter (more wires), lower power density than PM.

Property	DC	IM	PM	SRM
Efficiency	Low	Medium	Highest	Good
Power density	Low	Medium	Highest	Good
Cost	Medium	Low	High (magnets)	Lowest
Control	Simplest	Complex (FOC)	Complex	Moderate
Ruggedness / maint.	Brushes	Rugged	Good	Most rugged
Max speed	Low	High	Medium	Highest

5. Power Electronic Converters – DC/DC Converters

A DC/DC converter changes the DC voltage level (battery to inverter, battery to auxiliary loads, fuel cell / ultracap interface). It uses a high-speed switch (MOSFET / IGBT) operated with duty ratio $D = t_{on} / T$.

(a) Buck converter (step-down)

$$V_o = D\,V_{in} \quad (0 < D < 1)$$

When the switch S is ON, the inductor current rises and energy goes to the load; when OFF, the diode D freewheels

and L discharges. Output ripple is reduced by L and C. Used for 12 V / 48 V auxiliary supplies.

(b) Boost converter (step-up)

$$V_o = V_{in} / (1 - D)$$

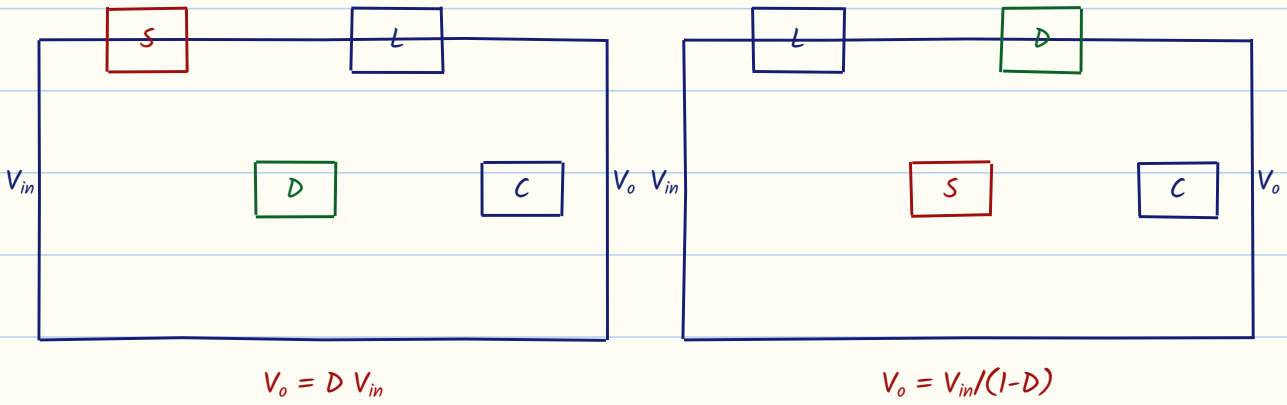
S ON: inductor stores energy from the source. S OFF: inductor emf adds to V_{in} and charges the output capacitor through the diode. Used to raise battery voltage to the higher DC-link voltage of the inverter.

(c) Buck-boost converter

$$V_o = - V_{in} D / (1 - D)$$

Gives output either lower ($D < 0.5$) or higher ($D > 0.5$) than input with **reversed polarity**. Inductor current: $\Delta I_L = V_{in} D T / L$ (CCM).

Buck (step-down) Boost (step-up)



$V_o = D V_{in}$ $V_o = V_{in}/(1-D)$

(d) Isolated DC/DC converter

When **galvanic isolation** between battery and the vehicle chassis / other bus is needed (safety, large voltage ratio), a **high-frequency transformer** is used. Types: **flyback**, **forward**, **push-pull**, **half-bridge** and **full-bridge**. The DC is first chopped (inverted) at 20-100 kHz, stepped up/down by the transformer, then rectified and filtered.

Isolated DC/DC: galvanic isolation through high-frequency transformer

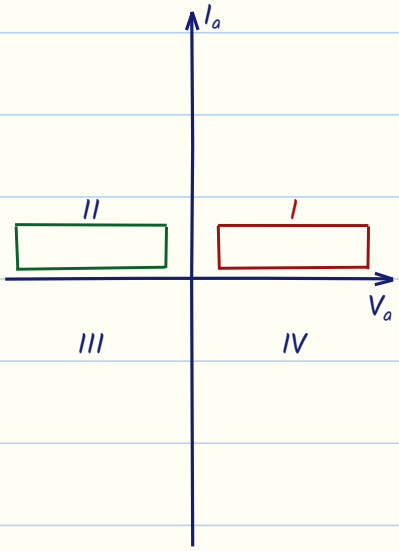


$V_o = D\,(N_2/N_1)\,V_{in}$ (forward / bridge type)

- Dual-active-bridge (DAB) is a bidirectional isolated converter - popular for onboard chargers and battery - DC link interface.
- Advantages of HF transformer: small size and weight, isolation, multiple outputs; losses: core and winding loss, leakage inductance spikes (need snubber).

6. Two-Quadrant Chopper and Switching Modes

A DC motor drive needs power flow both **from battery to motor (motoring)** and **motor to battery (regeneration)**. A **two-quadrant chopper (Class C)** has two switches S_1, S_2 with anti-parallel diodes D_1, D_2, forming a half-bridge leg. Terminal voltage stays positive and current may be positive or negative.



Quadrant I : forward motoring (S_1 chopping, buck mode)

Quadrant II : forward regeneration (S_2 chopping, boost mode)

Two-quadrant chopper: $V_a > 0$ always, I_a can be + (motor) or − (regen).

Operating modes

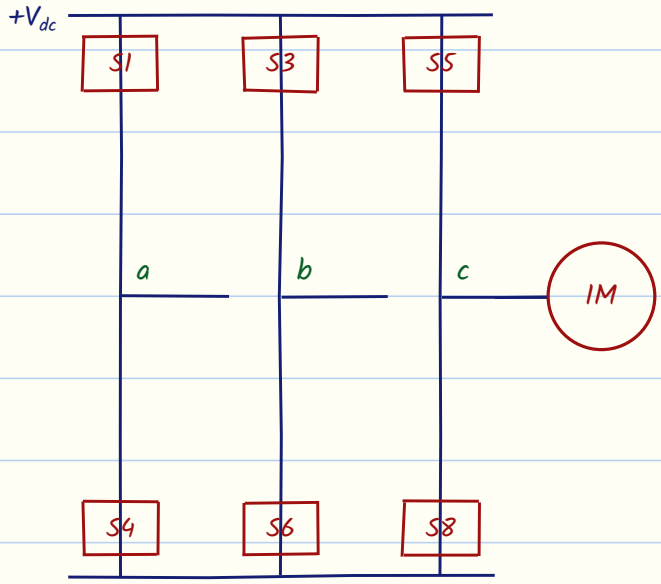
- **Motoring (buck mode)**: S_1 is switched with duty D, S_2 off. Mode 1: S_1 ON, current flows from battery through motor. Mode 2: S_1 OFF, D_2 freewheels. $V_a = D\ V_b$.
- **Regeneration (boost mode)**: S_2 is switched, S_1 off. Mode 1: S_2 ON, motor armature (back emf $E > V_b$-side) stores energy in inductance. Mode 2: S_2 OFF, current flows

through D_1 into the battery. $V_b = V_a/(1 - D)$.

- **Switching modes of control**: (i) constant-frequency **PWM** (vary t_{on}); (ii) variable-frequency (constant t_{on}); (iii) current-limit control. Dead time is provided between S_1 and S_2 to avoid shoot-through.

7. AC Drives - PWM and Current Control

An AC drive (IM or PMSM) is fed by a **three-phase voltage-source inverter (VSI)** from the DC bus. PWM produces variable-voltage variable-frequency (VVVF) output.



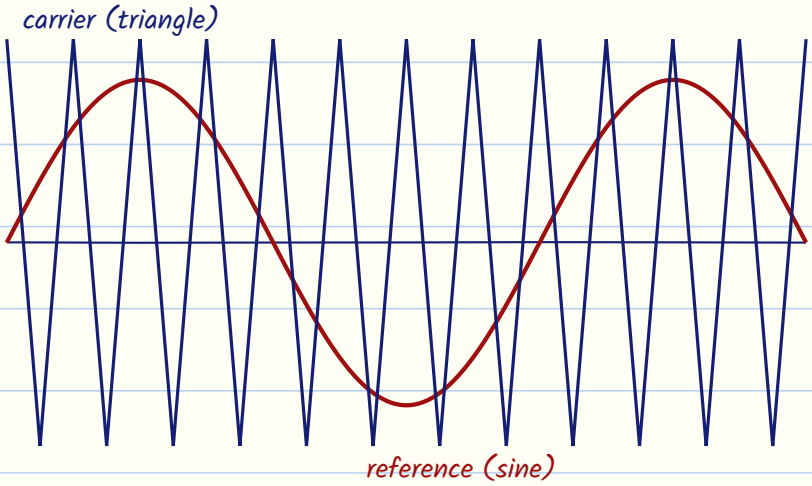
Three-phase voltage source inverter

6 IGBT / MOSFET + diodes
PWM gives variable V and f
Feeds IM or PMSM from battery
Bidirectional: regen possible

(a) Sinusoidal PWM (SPWM)

A sinusoidal reference (modulating) wave at the required output frequency is compared with a high-frequency triangular carrier. Where the reference exceeds the carrier the upper switch is ON. Amplitude modulation index m_a =

$V_{ref}/V_{carrier}$; carrier ratio $m_f = f_c/f_{ref}$. Fundamental line-line rms = $0.612\ m_a\ V_{dc}$. Over-modulation ($m_a > 1$) gives up to square wave with lower utilisation of harmonics.



Sine PWM:

$v_{ref} > v_{tri}$ -> upper switch ON
$v_{ref} < v_{tri}$ -> lower switch ON
$m_a = V_{ref}/V_{tri}$
$V_{LL,rms} = 0.612\ m_a\ V_{dc}$
($0 < m_a <= 1$, linear)

(b) Space-vector PWM (SVPWM)

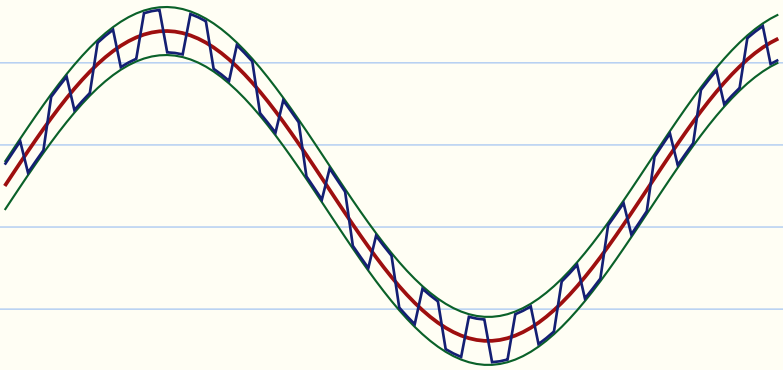
- Treats the three phase voltages as one rotating reference vector built from the 8 inverter switching states (6 active + 2 zero vectors).
- About **15 % higher DC bus utilisation** than SPWM and lower harmonic distortion, easy digital implementation - preferred in EVs.

(c) Current control methods

- **Hysteresis (bang-bang)** current control - the phase current is kept within ±h of the reference by switching the inverter leg when the band is crossed. Fast, simple;

frequency not fixed.

- **Ramp-comparison (PWM) current control** – the current error passes through a PI controller and is compared with the carrier. Fixed frequency, better harmonic spectrum.
- **Synchronous-frame PI control** – currents transformed to d-q axes become DC quantities, controlled by PI regulators (used with FOC).



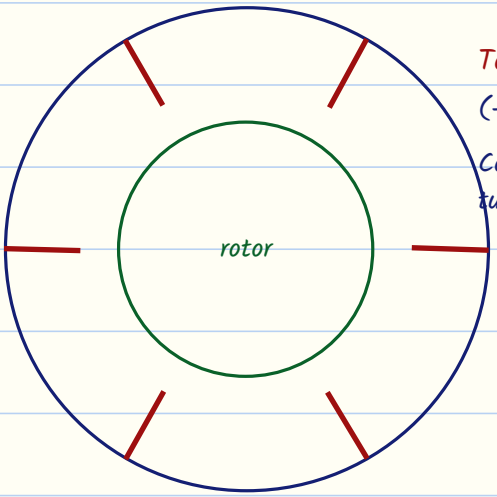
Hysteresis current control

actual current i stays inside band $i_{ref} \pm h$ (green). Simple, fast, robust but switching frequency varies.

8. Switched Reluctance Machine Drives

An SRM needs a converter that can energise each phase independently in the right rotor position. Common converter: **asymmetric half-bridge** per phase (two switches + two diodes). Switching both ON applies +V; both OFF applies -V (demagnetise); one OFF with diode conduction gives zero voltage (freewheel).

Phase winding with asymmetric half-bridge:



Torque $T = \frac{1}{2} i^2 (dL/d\theta)$

(+ve when $dL/d\theta > 0$: rising L)

Control variables: turn-on angle θ_{on}, turn-off angle θ_{off} and current / voltage.



6/4 SRM

(a) Voltage control

- The phase voltage (average) is controlled by PWM duty ratio or by choosing **turn-on and turn-off angles** (θ_{on}, θ_{off}). At high speed the back emf is large, current is naturally limited and **single-pulse (angle) control** is used.
- Simple, but the current peak is not directly controlled, so torque ripple and peak currents are larger at low speed.

(b) Current control

- Used at low speed and starting, where the current would otherwise rise too high. The phase current is regulated by **hysteresis** or PWM current controller around a reference I_{ref}, produced from the torque command (torque sharing between phases reduces ripple).
- Gives good torque control and protects switches; needs

current sensors in each phase.

- Overall: speed loop (outer) -> torque / current command -> current loop (inner) -> converter; rotor position feedback from encoder or sensorless methods.

Important questions: (1) Compare IM, PM and SRM for EV. (2) Buck and boost converter operation with equations. (3) Two-quadrant chopper. (4) Sine PWM and space vector PWM. (5) SRM voltage and current control.

INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT III

Hybrid and Electric Drive Trains

I. Introduction to Hybrid Electric Vehicles

A **Hybrid Electric Vehicle (HEV)** combines two or more energy sources / power sources - normally an **internal combustion engine (ICE)** and an **electric motor with battery** - to drive the vehicle. The engine can be run at its most efficient point and the electric path recovers braking energy.

Concept of hybridisation

- **Micro hybrid** - stop-start only. **Mild hybrid** - small motor assists engine (BAS, 48 V). **Full hybrid** - can run in pure electric mode (Prius). **Plug-in hybrid** - large battery, grid charging.
- Fuel economy gain: 10-15 % (mild) to 30-50 % (full hybrid) in city driving.

History of hybrid vehicles

- 1830s-1890s: first electric carriages; 1900: **Lohner-Porsche** series hybrid with in-wheel hub motors.

- 1900-1920: petrol cars (Ford Model T, electric starter 1912) displaced EVs because of cheap oil and long range.
- 1970s: oil crises renewed interest; 1990s: California ZEV rules. 1997: **Toyota Prius** (Japan) - first mass-produced hybrid. 1999: **Honda Insight** in USA. 2000s: Ford Escape, Lexus hybrids. 2010: **Chevrolet Volt** (plug-in), **Nissan Leaf** (BEV). Now: Tesla, BYD, Tata Nexon EV etc.

Social importance

- Reduces dependence on imported petroleum (energy security) and fuel bills.
- Lowers CO_2, NO_x, HC and particulate emissions - better urban air quality and less global warming.
- Less noise; creates new technology industry and jobs; supports renewable energy via grid charging.

Impact of modern drive trains on energy supplies

- Transport uses about 25-30 % of world energy and is 95 % oil-based. Higher efficiency drive trains reduce oil demand; each 1 % fuel saving across the fleet is a large barrel-per-day saving.
- PHEV / EV shift part of transport energy to the **electric grid**: increases electricity demand but allows use of coal, gas, nuclear, solar and wind; **well-to-wheel efficiency** of

EV (about 25-35 %) is better than ICE (about 12-20 %).

- Smart charging and vehicle-to-grid (V2G) can support grid stability; battery material (lithium, cobalt) supply and recycling become new issues.

2. Hybrid Traction and Electric Traction

Aspect	Hybrid traction	Electric traction
Energy source	ICE (fuel) + battery / motor	Battery / fuel cell / ultracap only
Emission	Low, some tail-pipe	Zero tail-pipe
Range	Long (fuel tank)	Limited by battery
Refuelling	Petrol + optional plug	Charging / H_2 filling
Complexity	High (2 power plants)	Simple drive train
Examples	Prius, Insight, Volt	Leaf, EVI, Tata Nexon EV

Hybrid traction uses ICE and electric machine in series, parallel or combined arrangement. **Electric traction** uses only electric machines with energy from battery, fuel cell or supercapacitor; also used in electric trains and trams.

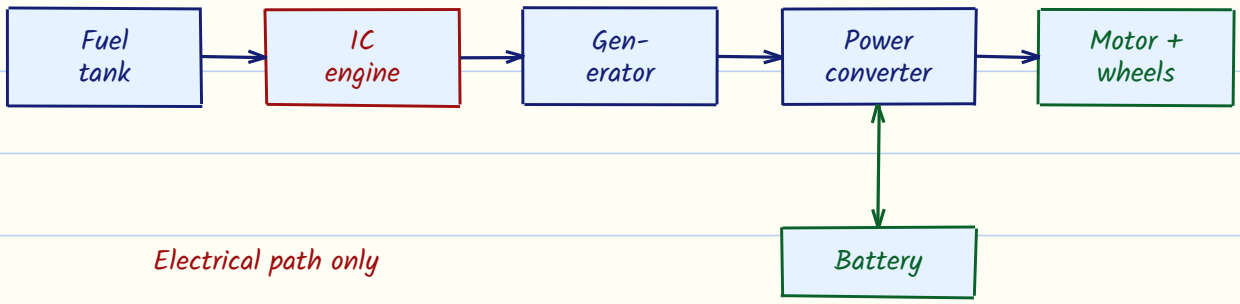
3. Hybrid and Electric Drive Train Topologies

(a) Series hybrid

The ICE drives a **generator**; electric power goes to the battery and traction motor. Only the motor is coupled to the wheels. The engine speed is independent of vehicle speed, so

it can run at optimum efficiency.

Series hybrid: engine NOT mechanically connected to wheels

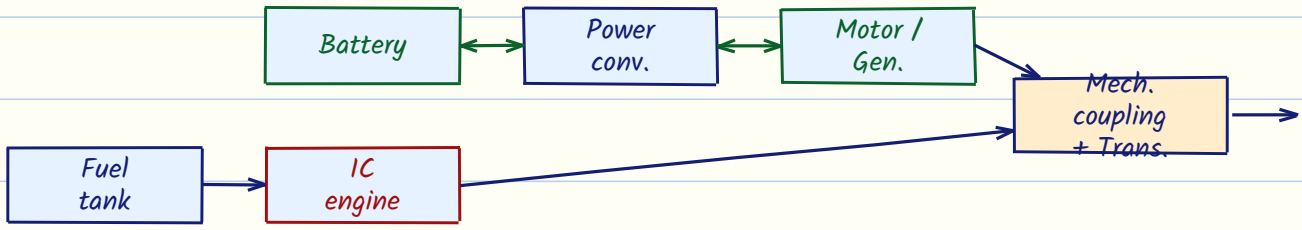


- + Simple control, engine at best efficiency point, no clutch or gear box, flexible layout.
- - Energy is converted twice (mech -> elec -> mech), so lower efficiency at highway speed; needs three machines of full rating (engine, generator, motor). Suitable for city buses, locomotives, range extenders (Volt in extended mode).

(b) Parallel hybrid

Both ICE and electric motor are connected to the transmission through a **mechanical coupling**; they can drive wheels together or separately. The motor also acts as generator / starter.

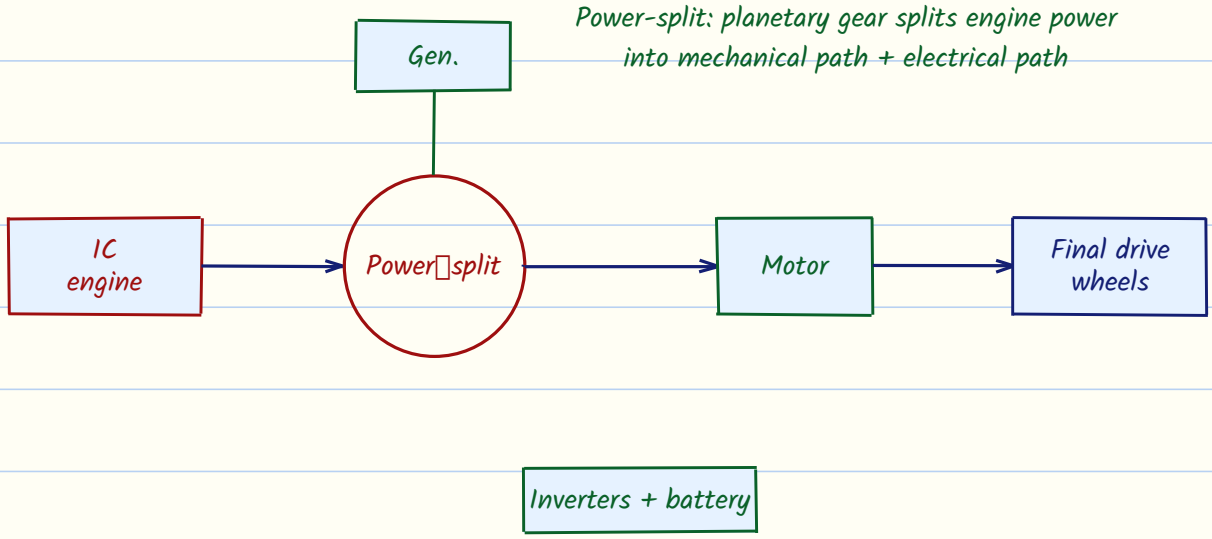
Parallel hybrid: engine and motor both drive the wheels



- + Smaller motor and battery, only one electric machine, high efficiency at highway, no energy conversion when engine drives.
- - Complicated mechanical coupling and control, engine speed linked to wheels so not always at best point. Ex: Honda Insight (IMA), mild hybrids.

(c) Series-parallel (power-split) hybrid

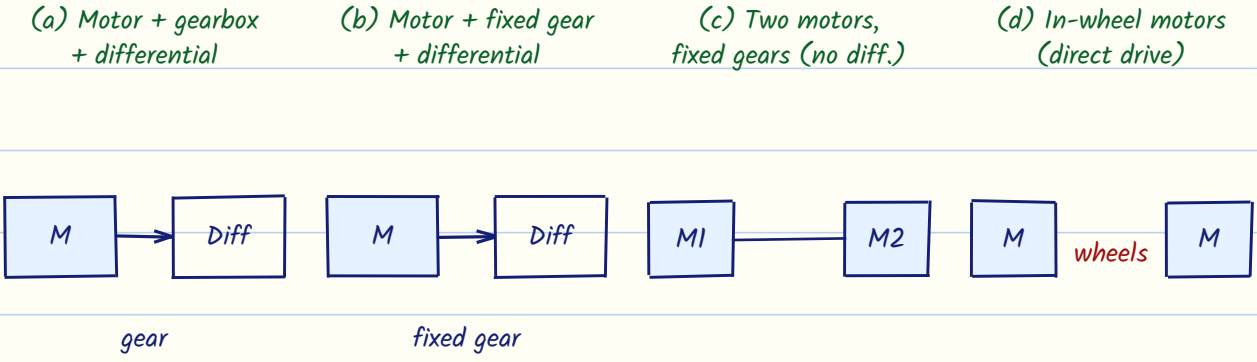
A **planetary gear (power split device)** divides engine power into a mechanical path (to wheels) and an electrical path (generator -> motor). Gives the advantages of both: engine speed is decoupled from wheel speed (like CVT) and direct mechanical drive. Ex: **Toyota Prius**.



(d) Complex hybrid

Similar to series-parallel but the generator and motor can switch roles using more power electronics (bi-directional). More flexible but costly. Ex: Lexus RX400h (four-wheel drive).

Electric (BEV) drive train topologies



- (a) is the conventional layout; (b) removes gearbox - single-speed reduction; (c) two motors give electronic differential; (d) **in-wheel / hub motors** remove all

transmission but raise unsprung mass.

4. Power Flow Control and Energy Efficiency Analysis

Power flow control means deciding, at each instant, how much power comes from engine, motor/generator and battery so that the **demand** at the wheel is met with minimum fuel and the battery SOC is kept within limits.

Operating modes of a series hybrid

- **Pure electric (low speed)** - battery -> motor -> wheels, engine off.
- **Engine alone (cruise)** - engine -> generator -> motor -> wheels, battery idle.
- **Hybrid (acceleration)** - engine-generator and battery both supply the motor.
- **Battery charging** - engine -> generator -> battery (vehicle stopped or light load).
- **Regenerative braking** - wheels -> motor (as generator) -> battery.
- **Combined**: charging while driving; **stand still** - engine off.

Operating modes of a parallel hybrid

- Motor only (start / low load); engine only (highway); engine + motor (peak power); engine drives vehicle and charges battery through motor-generator; regenerative

braking; stop-start.

Power balance equations

$$P_{wheel} = P_{engine}\,\eta_{m1} + P_{motor}\,\eta_{m2}$$
$$P_{batt} = (P_{motor} / \eta_m) - (P_{gen}\,\eta_g)$$

Supervisory (energy management) controller chooses the split by rules (thermostat / power-follower) or by optimisation to keep ICE in its high-efficiency zone and SOC between 40 % and 80 %.

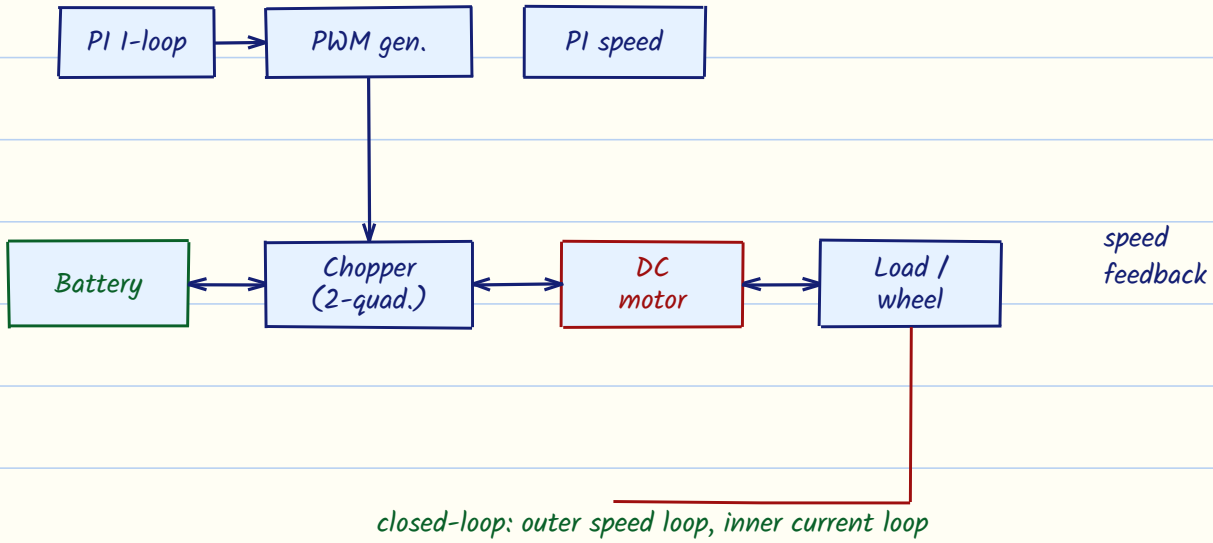
Energy efficiency analysis

- Efficiency of a path = product of component efficiencies. Series path: $\eta_{ICE}\,\eta_{gen}\,\eta_{conv}\,\eta_{batt}\,\eta_{motor}\,\eta_{trans}$ (low); parallel direct path: $\eta_{ICE}\,\eta_{trans}$ (high).
- Fuel saving comes from: (i) engine downsizing, (ii) running engine near best brake specific fuel consumption (BSFC) point, (iii) engine off at stop, (iv) regeneration recovers 15–30 % of braking energy.
- Example: series path with η_{ICE}=0.35, gen 0.92, conv 0.96, motor 0.92, gear 0.97 gives 0.35×0.92×0.96×0.92×0.97 = **0.276 (27.6 %)**, while the direct parallel path gives 0.35×0.95 = 0.33 (33 %).

5. Configuration and Control of Motor Drives

(a) DC motor drives

Separately excited DC motor: $V_a = E + I_a R_a$, $E = K_e \varphi \omega$, $T = K_t \varphi I_a$. Speed is controlled by **armature voltage** (below base speed, constant torque) and by **field weakening** (above base speed, constant power).



- A DC-DC chopper (two-quadrant or four-quadrant H-bridge) supplies the armature from the battery. Control: inner current loop (fast) limits torque; outer speed loop gives reference current.
- + Simple control, good low speed torque; - brushes, commutator wear, low speed limit. Now largely replaced.

(b) Induction motor drives

- Battery -> **three-phase PWM inverter** -> IM. Configured for V/f control or **field-oriented control** with encoder /

sensorless speed estimation.

- FOC: i_{sd} (flux) and i_{sq} (torque) regulated by PI controllers in the rotating frame; field weakening above base speed using flux reference $\psi_r^* \propto 1/\omega$.

(c) Permanent magnet motor drives

- Inverter-fed PMSM with rotor position sensor (resolver). **Maximum torque per ampere (MTPA)** below base speed and **flux weakening** (negative i_d) above base speed. BLDC drive uses six-step commutation with Hall sensors.

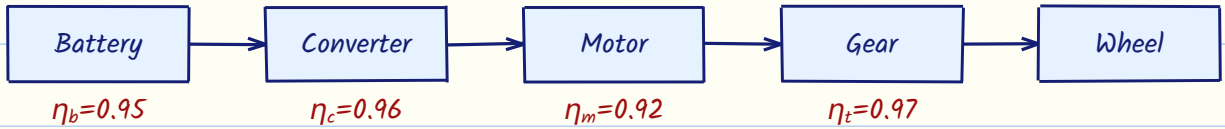
(d) Switched reluctance motor drives

- Asymmetric half-bridge converter per phase, rotor position feedback. Low speed: current (hysteresis) control; high speed: single-pulse angle control. Torque sharing function reduces ripple.

6. Drive System Efficiency

The **overall drive system efficiency** from battery to wheel is the product of the efficiencies of converter, motor and transmission. It is not constant - it depends on torque and speed (efficiency map).

Drive system efficiency = product of efficiencies in the chain



$\eta = 0.95 \times 0.96 \times 0.92 \times 0.97 = 0.81$ (81 %)

- Motor efficiency map: peak 90-96 % in middle speed and 50-80 % load; efficiency falls at very low torque (fixed losses) and at very high speed / torque (copper and iron losses).
- Losses: **copper loss** I^2R, **iron loss** (hysteresis + eddy), **mechanical** (friction, windage), **switching and conduction loss** in converter, gear loss.
- Improve efficiency by: better design (IPM), field weakening, high-efficiency SiC devices, optimal gear ratio so the motor operates in its sweet spot, using regeneration.

$\eta_{regen} = \eta_t \eta_m \eta_c \eta_b \approx 0.7 - 0.8$ (braking energy returned to battery

Important questions: (1) Explain history and social importance of HEV. (2) Series, parallel and series-parallel topologies with figures. (3) Power flow control in hybrid modes. (4) Control of DC / IM / PM / SRM drives. (5) Drive system efficiency.

INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT IV

Electric and Hybrid Vehicles - Case Studies

I. Series and Parallel Hybrids - Operating Strategies

The case studies in this unit are based on the two basic hybrid architectures (series, parallel) and their combinations. Operation is described by two strategies: **charge sustaining** and **charge depleting**.

Series hybrid

- Engine -> generator -> electric motor; battery buffers power. Wheels driven only by the motor. Ex: Chevrolet Volt (in extended range), diesel-electric locomotives and hybrid buses.

Parallel hybrid

- Engine and motor both couple mechanically to wheels. Ex: Honda Insight / Civic hybrid (IMA), many mild hybrids.

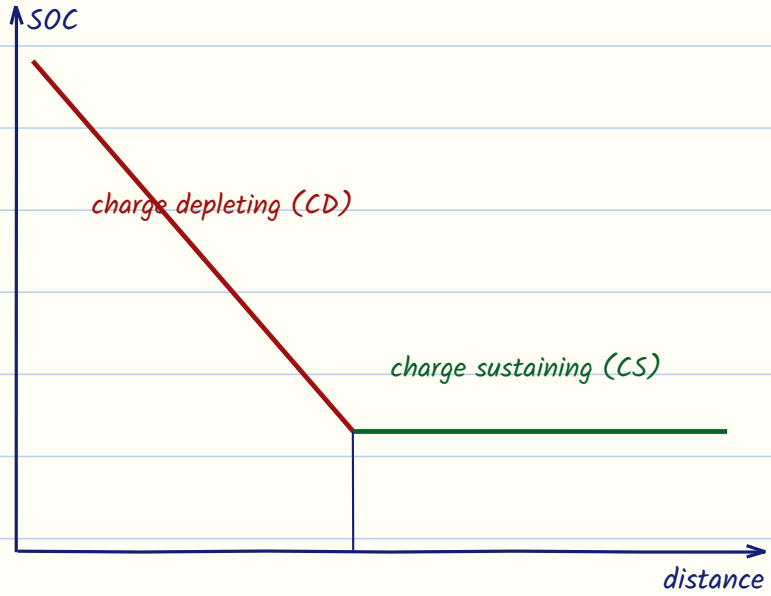
Charge sustaining (CS) mode

- The battery SOC is **maintained within a narrow band** about a target (say 50-60 %). The engine supplies the average power; the battery only supplies peaks and absorbs

regenerative energy. Net battery energy change over the trip is about zero; all energy comes from fuel. Used in conventional HEVs such as the Prius and Insight.

Charge depleting (CD) mode

- The vehicle uses **stored battery energy** (from the grid) to propel, SOC falls from high to a lower limit. This is the **all-electric range (AER)** of a PHEV / EREV. After reaching the lower SOC limit the vehicle switches to CS mode (Volt: about 40 km EV, then engine runs).

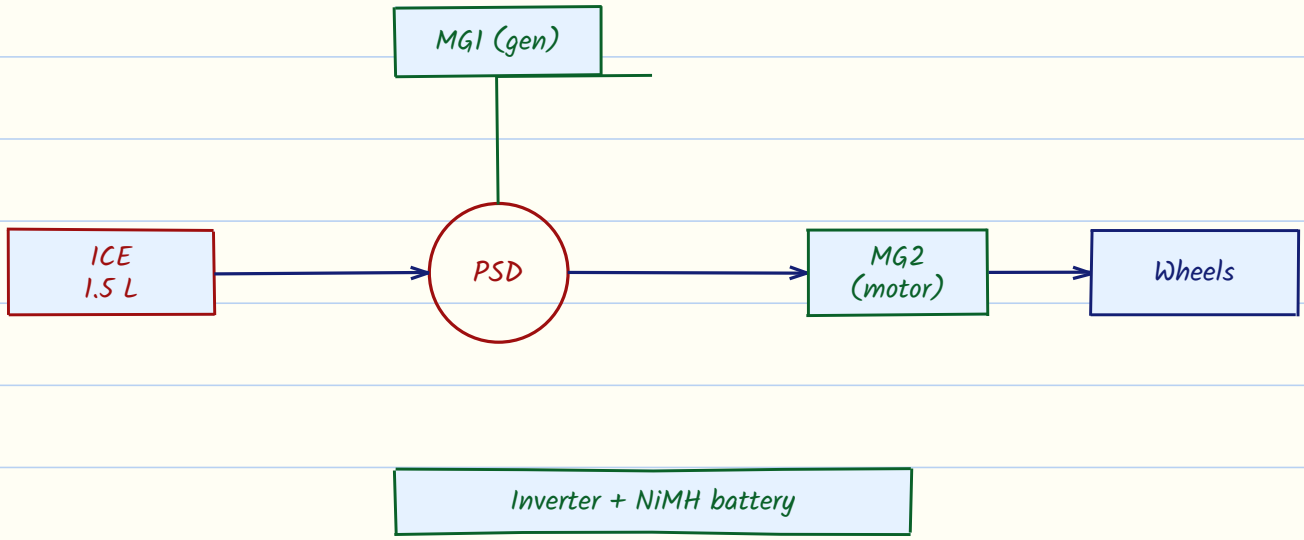


CD : battery energy used up, SOC falls (EV mode).

CS : SOC held near a set value; fuel supplies all the energy.

2. Hybrid Vehicle Case Study - Toyota Prius

The **Toyota Prius** (1997 Japan, 2000 world) is the world's first mass-produced hybrid and best-selling HEV. It is a **series-parallel (power-split)** hybrid with the **Toyota Hybrid System (THS / HSD)**.



- **Engine**: Atkinson-cycle petrol engine - 1.5 L, about 57 kW (2nd gen); later 1.8 L, about 73 kW. Atkinson cycle gives high efficiency but lower power density, compensated by the motor.
- **Power split device**: planetary gear. Engine -> planet carrier, MG1 (generator) -> sun gear, MG2 (traction motor) and wheels -> ring gear. Acts like an **electronic CVT**; no clutch, no conventional gearbox.
- **Motors**: MG1 and MG2 are PM synchronous (AC) machines; MG2 about 50 kW (2nd gen), 60 kW (3rd gen), fed by an inverter; boost converter raises battery voltage (201.6 V -> up to 500 V) in later models.
- **Battery**: sealed NiMH, about 201.6 V (168 cells in 2nd gen), about 6.5 Ah; SOC kept between 40-80 %.
- **Modes**: starts and low speed on motor only; normal driving

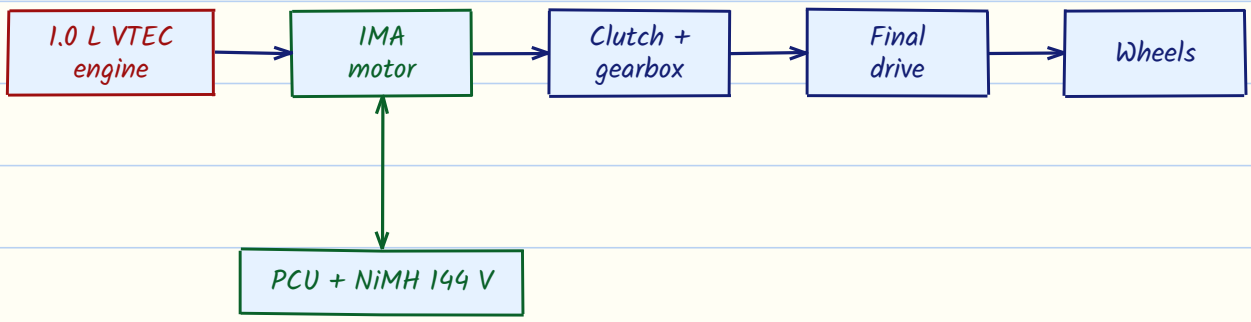
engine + MG2, MG1 charges the battery; full throttle engine + battery power; braking - regeneration by MG2 (+ hydraulic brake blending); engine auto stop at standstill.

- Fuel economy about 4-5 L/100 km (about 20 km/L); emissions much lower than comparable ICE cars.

3. Hybrid Vehicle Case Study - Honda Insight

The first-generation **Honda Insight (1999)** is a two-seat **parallel mild hybrid** using the **Integrated Motor Assist (IMA)** system. The thin permanent-magnet DC brushless motor is mounted directly on the engine crankshaft, between engine and transmission, so no separate clutch for it.

Motor sits between engine and transmission (IMA)



- **Engine**: 1.0 L, 3-cylinder, VTEC-E lean burn, about 50 kW (67 hp). **Motor**: about 10 kW (13 hp) assist; works as starter, assist and generator (regeneration).
- **Battery**: NiMH, 144 V, 6.5 Ah (120 cells) under the rear

floor. Transmission: 5-speed manual or CVT.

- **Light weight** (aluminium body about 850 kg) and excellent aerodynamics (C_D = 0.25) help to give about 3.4 L/100 km (very high economy).
- Modes: motor assists at acceleration; cruise by engine alone; regeneration on deceleration; **idle stop**; lean-burn cylinder deactivation.
- Merits: simple, cheap, only one electric machine; demerit: cannot run in pure-EV mode, assist motor is small.

4. Hybrid Vehicle Case Study - Chevrolet Volt

The **Chevrolet Volt (2010-11)** is a **plug-in extended-range electric vehicle (EREV)**. It is mostly an electric car with a small petrol engine used only for generating power (or assisting at high speed).

Chevrolet Volt operating modes

1. EV mode (CD)	battery -> motor -> wheels; engine off
2. Extended range (CS)	engine -> generator -> motor; battery holds SOC
3. High-speed assist	engine also helps drive through planetary gears
4. Regeneration	motor -> battery while braking

- **Battery**: T-shaped Li-ion pack, about 16 kWh (about 10.5 kWh usable), 288 cells, liquid-cooled; **EV range about 35-40 miles (about 60 km)**; charged from a 120 / 240 V

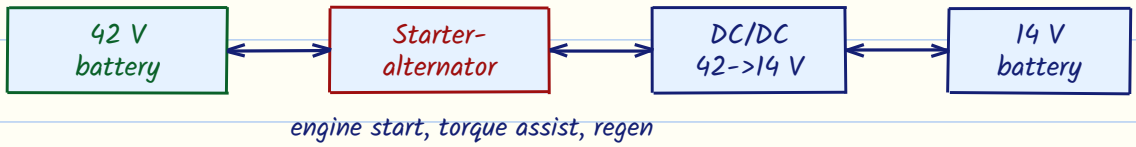
outlet in about 4-10 hours.

- **Motors**: main traction motor about 111 kW (150 hp), 370 N-m; second motor/generator about 55 kW. Engine: 1.4 L petrol, about 63 kW. Planetary gear and clutches allow engine to assist directly at higher speed (2nd gen improved).
- Total range about 600 km. Charge depleting first, then charge sustaining. Typical commuter drives entirely on grid electricity.

5. 42 V System for Traction Applications

Modern cars need more and more electrical power (steering, air-conditioning, valves, heating, infotainment). The old **14 V (12 V battery)** system can supply only about 2-3 kW because of high current. The industry proposed the **42 V PowerNet** (36 V battery, 42 V charging).

Dual-voltage system: heavy loads on 42 V, lamps / electronics on 14 V



engine start, torque assist, regen

EPS, A/C compressor, valve actuators, heated catalyst on 42 V bus

- Higher voltage -> for the same power the current is 3

times less, so thinner copper wire, smaller connectors, less I^2R loss; allows 5-10 kW generators.

- Enables **starter-alternator (integrated starter generator, ISG / BSG)**: engine start-stop, torque assist, regeneration, electric power steering, electric water pump, electromechanical valve actuators, electric A/C compressor, electrically heated catalyst.
- Issues: arcing at switches above about 30 V (needs solid-state switches), new bulbs / motors, safety below 60 V (not hazardous - below the touch-voltage limit), dual-voltage system complexity. It was later replaced by the **48 V** mild hybrid standard.

6. Lightly Hybridised Vehicles and Low-Voltage Systems

- **Micro hybrid**: only stop-start (belt starter), 12 V; saves 3-8 % fuel.
- **Mild hybrid**: 42 / 48 V system with belt-driven starter-generator (BSG) or crank-mounted ISG of 5-15 kW; provides stop-start, regenerative braking and light **torque assist**; **cannot drive the vehicle by motor alone**. Ex: Honda IMA, GM eAssist, Suzuki SHVS, Maruti Ciaz / Ertiga (SHVS), Mercedes EQ Boost.
- Fuel economy gain: 10-20 %; cost is modest compared to

full hybrids; small battery (0.3-1 kWh, Li-ion or supercap).

- **Low voltage (below 60 V DC)** is regarded as safe for people (no insulation hazard) and cheaper to build.

Type	Voltage	Motor power	Electric-only drive	Fuel saving
Micro	12 V	2-3 kW	No	3-8 %
Mild	42/48 V	5-15 kW	No	10-20 %
Full	200-300 V	30-60 kW	Yes	30-50 %
Plug-in	300+ V	60-110 kW	Yes (long)	50 %+

7. Electric Vehicle Case Studies

(a) GM EV1

- General Motors **EV1 (1996-1999)**, the first modern mass-produced electric car (leased in California / Arizona). Two-seat, very streamlined (C_D = 0.19), aluminium structure.
- **Motor**: 3-phase AC induction motor about 102 kW (137 hp) with IGBT inverter; single-speed reduction gear; regenerative braking.
- **Battery**: Gen 1 lead-acid 16.5 kWh (range 55-95 miles); Gen 2 (1999) NiMH 26.4 kWh (range about 100-140 miles). Top speed limited to 80 mph (about 130 km/h);

0-60 mph in about 8 s. Inductive (Magne Charge) charging.

- Program ended in 2002-03 (leases not renewed); important as the technology demonstrator that inspired Volt and later EVs.

(b) Nissan Leaf

- **Nissan Leaf (2010)** - best-selling BEV for years. 5-door hatchback front-wheel drive.
- **Motor**: 80 kW (107 hp) PM synchronous AC motor, 280 N-m, single-speed reduction.
- **Battery**: 24 kWh (2010) laminated-cell Li-ion pack (48 modules, about 360 V) under floor; range about 117 km EPA (about 175 km NEDC); later 30 / 40 / 62 kWh versions.
- Charging: 3.3 / 6.6 kW AC onboard charger (8 h on 240 V), **CHAdeMO DC fast charge** 80 % in about 30 min. Also supports vehicle-to-home (V2H).

(c) Mitsubishi i-MiEV

- **i-MiEV (2009)** - based on the small "i" kei car; rear-mounted motor and rear-wheel drive; low floor and compact.
- **Motor**: 47 kW PM synchronous, 180 N-m. **Battery**: 16 kWh

Li-ion (88 cells, 330 V) under the floor; range about 160 km (JC08), about 100 km real.

- Charging: 15 h on 100 V, 7 h on 200 V, 80 % in 30 min with CHAdeMO quick charge. Top speed 130 km/h.

Parameter	GM EV1	Nissan Leaf	i-MiEV
Year	1996 / 99	2010	2009
Motor	AC induction, 102 kW	PMSM, 80 kW	PMSM, 47 kW
Battery	Lead-acid / NiMH	Li-ion 24 kWh	Li-ion 16 kWh
Range	~100-225 km	~117-175 km	~100-160 km
Top speed	130 km/h	145 km/h	130 km/h

8. Hybrid Electric Heavy-Duty Vehicles

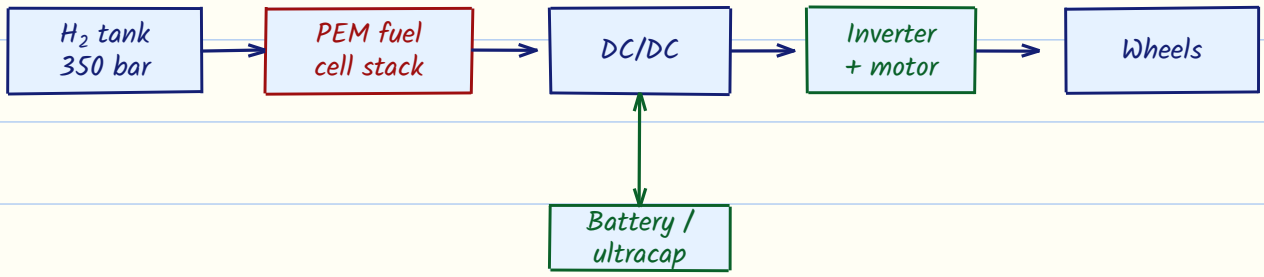
Trucks, transit buses, refuse trucks and military vehicles have stop-go duty cycles and large mass, so **regenerative braking and engine downsizing** give large fuel savings (20-40 %).

- **Series hybrid buses** (Orion VII, Alexander Dennis Enviro 400H): diesel engine drives generator; large battery / ultracapacitor and wheel-hub or central traction motor. Engine runs at constant best-efficiency point.
- **Parallel hybrid trucks** (Eaton, Volvo I-SAM, Kenworth T270): motor-generator between engine and automated transmission; torque assist and regeneration.

- **Hydraulic hybrids** (UPS / refuse trucks) store energy in a pressurised accumulator instead of batteries.
- Challenges: heavy and costly battery, thermal management, high power converters (200-600 V, 100-250 kW), duty cycle dependency; use ultracapacitors for peaks.

9. Fuel Cell Heavy-Duty Vehicles

Fuel-cell hybrid heavy-duty power train



- **PEM fuel cell** (proton exchange membrane): anode $H_2 \rightarrow 2H^+ + 2e^-$; cathode $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$. Electric output about 0.6-0.7 V per cell, efficiency 40-60 %; the only exhaust is water.
- Hydrogen stored at 350 / 700 bar in carbon-fibre tanks; refuelling in 5-10 minutes; range 300-500+ km, good for buses and trucks needing long range and heavy payload.
- Fuel cell is slow to respond, so it is **hybridised with a battery / ultracapacitor** for peak power and regeneration. Ex: Ballard-powered city buses, Mercedes Citaro FuelCELL,

Toyota Mirai (cars), Hyundai XCIENT Fuel Cell truck.

- + Zero tail-pipe emission, long range, fast refuelling. - High cost (platinum catalyst), need hydrogen production / distribution, durability and cold-start issues.

Important questions: (1) Charge sustaining vs charge depleting. (2) Toyota Prius / Honda Insight / Chevrolet Volt systems. (3) 42 V system. (4) GM EVI, Nissan Leaf and Mitsubishi i-MiEV. (5) Heavy-duty hybrid and fuel cell vehicles.

INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT V

Electric and Hybrid Vehicle Design

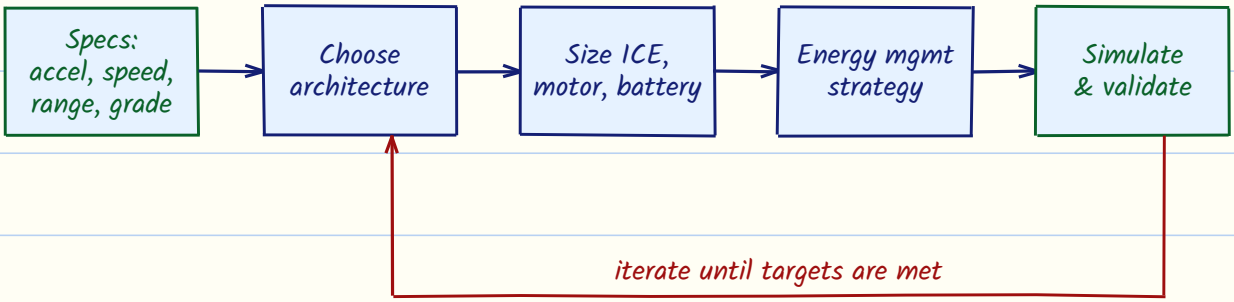
I. Introduction to Hybrid Vehicle Design

Designing a hybrid / electric vehicle means choosing the **architecture** and then **sizing every component** (engine, motor, power electronics, battery, drive train) so that the vehicle meets performance targets with the lowest fuel use, cost and weight.

Design requirements (specifications)

- **Acceleration**: 0-100 km/h in t_a seconds (e.g. 10 s). **Top speed** (e.g. 150 km/h). **Gradeability**: climb x % slope at given speed (e.g. 20 % at low speed, 5 % at 100 km/h).
- **Range** / fuel economy on standard drive cycles (**NEDC, WLTP, FTP-75, Indian IDC**), cruise range, emission limits.
- Weight, cost, package space, safety (high-voltage), life 8-10 years, comfort (NVH) and charging time.

Design procedure



- **Step 1** fix vehicle parameters (m, C_D, A_f, μ_{rr}, wheel radius); **Step 2** compute power demand for each requirement; **Step 3** size motor and ICE; **Step 4** size battery for power and energy; **Step 5** develop control strategy; **Step 6** simulate on drive cycles (MATLAB/Simulink, ADVISOR, Autonomie) and optimise.

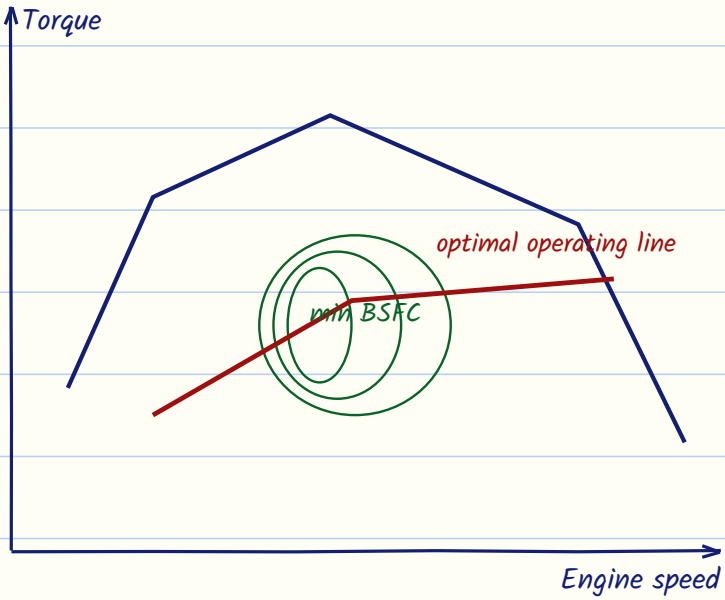
Degree of hybridisation

$$H = P_{em} / (P_{em} + P_{ICE})$$

$H = 0$ conventional car, $H = 1$ pure EV. Micro / mild hybrid: $H \approx 0.1$–0.2; full hybrid: 0.3–0.5; plug-in / EREV: 0.5–0.8.

2. Matching the Electric Machine and the ICE

The two power sources must complement each other: the **ICE** is good for long-duration, steady power at high load (but inefficient at low load); the **electric machine** is good for transients, low speed torque and regeneration.



Matching ICE and motor:
Operate ICE on the optimal line (best BSFC contours). Motor fills the gap between demand and ICE output and absorbs excess power.

Parallel hybrid – torque coupling

$$T_{out} = k_1 T_{ICE} + k_2 T_{EM}$$

- Torque coupler (gear / belt) adds torques; speeds are proportional: $\omega_{out} = \omega_{ICE}/k_1 = \omega_{EM}/k_2$. Used in Honda IMA, parallel hybrids. Motor and engine ratings chosen so that **ICE alone** satisfies cruising and mild gradients, motor adds in peak demands.

Speed coupling (series-parallel / power split)

$$\omega_{out} = k_1 \omega_{ICE} + k_2 \omega_{EM}$$

- Planetary gear adds speeds while torques are proportional, so the engine speed can be varied independently of the wheel speed (CVT action).

Matching rules

- **ICE power** is selected to meet steady-state demand: $P_{ICE} \geq P_{cruise\ at\ v_\{max\}}$ (or at max gradient and speed) / η_t, usually 50-70 % of the equivalent conventional engine.
- **Series hybrid**: engine-generator rating = average power demand (about 25-30 kW for a car) + battery charging power; motor must meet the **peak** demand alone.
- Operate ICE on the **optimal operating line** on the BSFC map (minimum fuel for each power). Choose gear ratios so that cruising lies near the line.
- Match maximum speeds: ICE max rpm / gear ratio ≥ vehicle top speed; motor base speed ≈ engine idle-to-mid-speed range.

3. Sizing of the Propulsion Motor

The motor must provide peak power for **acceleration**, continuous power for **top-speed cruising** and enough torque for **gradeability**. The rated power is the maximum of the three.

(a) Power for acceleration (0 to v_f in t_a s)

$$P_m = (\delta\, m / 2\, t_a)\,(v_f^2 + v_b^2) + (2/3)\, m\, g\, \mu_{rr}\, v_f + (1/5)\, \rho\, C_D\, A_f\, v_f$$

v_b = base speed (vehicle speed at motor base speed); v_f = final speed; δ = rotational mass factor (1.05).

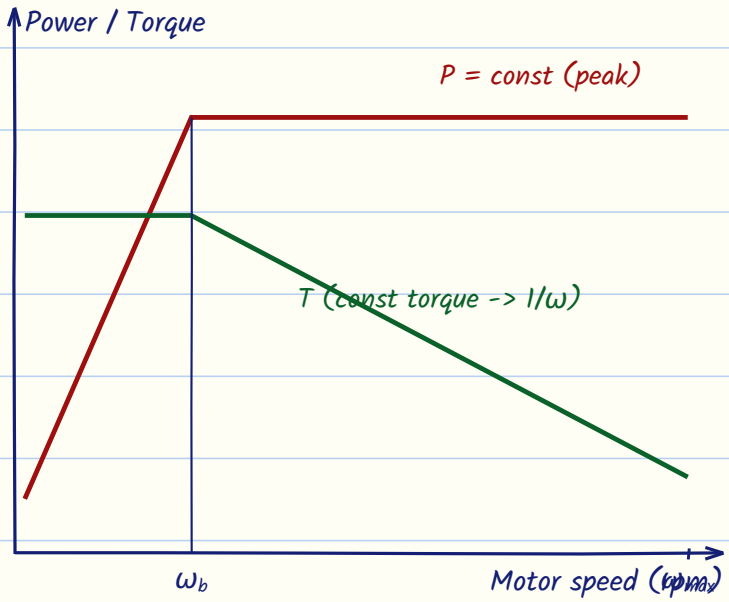
- Example: $m = 1500$ kg, $v_f = 100$ km/h = 27.8 m/s, $v_b = 50$ km/h = 13.9 m/s, $t_a = 10$ s, $\mu_{rr} = 0.012$, $\rho C_D A_f = 0.792$.
- Term 1 = $(1.05 \times 1500 / 20)(27.8^2 + 13.9^2) = 78.75 \times 966 \approx$ **76 kW**; Term 2 = $(2/3)(1500)(9.81)(0.012)(27.8) \approx 3.3$ kW; Term 3 = $0.2 \times 0.792 \times 27.8^3 \approx 3.4$ kW.
- $P_m \approx$ **83 kW** (peak). Continuous rating for steady 100-130 km/h cruise is much lower (15-25 kW); the motor is rated for peak with short-time overload (1.5-2×).

(b) Power for top speed and gradient

$$P_m = (v / \eta_t) [m g \mu_{rr} \cos\alpha + m g \sin\alpha + \tfrac{1}{2} \rho C_D A_f v^2]$$

(c) Torque and speed rating

- Base speed ω_b: motor rated torque $T_{rated} = P_{rated} / \omega_b$. Larger $x = \omega_{max}/\omega_b$ gives smaller T_{rated} for the same peak power (smaller motor and inverter) but demands good field weakening.
- Gear ratio from top speed: $G = \omega_{max} r / v_{max}$. Example: 9000 rpm (942 rad/s), $r = 0.3$ m, $v_{max} = 150$ km/h = 41.7 m/s -> $G = 942 \times 0.3/41.7 =$ **6.8 ≈ 7**.



Rated power from:
(1) acceleration time
(2) top speed (cruise)
(3) max gradient at speed
Take the LARGEST value.
Constant-power ratio $x = \omega_{max}/\omega_b$
typ. 3-4 (IM / IPM).

Remember: acceleration time normally decides the peak power; gradeability decides the maximum torque; top speed decides the continuous power.

4. Sizing of Power Electronics

- **Inverter current rating** from motor peak power: $I_{rms} = P_{peak} / (\sqrt{3}\, V_{LL} \cos\varphi\, \eta)$. Example: 83 kW, V_{LL} = 200 V, pf = 0.9, η = 0.95 -> I_{rms} = 83000/(1.732×200×0.9×0.95) = **280 A rms (about 396 A peak)**.
- **DC-link voltage** $V_{dc} \geq \sqrt{2}\, V_{LL}/m_a$ (e.g. 200 V line gives about 300-350 V bus) plus margin 20-30 % for regenerative overshoot; semiconductor rating = 1.5-2 × V_{dc} (e.g. 650 / 1200 V IGBT).
- **Switching device**: IGBT (high power), MOSFET (low voltage, <200 V), **SiC MOSFET** (high efficiency, high

frequency). Choose switching frequency 8-20 kHz; select gate driver and snubber.

- **DC-link capacitor** to absorb ripple current (film capacitor), sized for ripple $I_{c,rms}$ and voltage ripple.
- **Thermal design**: loss P_{loss} = conduction + switching; heat-sink / liquid cooling to keep junction below 125-150 °C; DC-DC converter sized for auxiliary load (2-3 kW, 12 V).
- Include protection: over-current, over-voltage, short circuit, temperature, isolation monitoring.

5. Sizing of the Drive System

- **Transmission**: single-speed reduction (G = 6-10) for EVs; two-speed for high-performance or heavy EV; CVT / planetary in HEVs. Efficiency of gear 95-98 %.
- Check wheel torque: $T_{wheel} = T_m\, G\, \eta_t$; tractive force $F = T_{wheel}/r$ must exceed required F_{te} at start, and be below adhesion limit $\mu_a N$.
- Select final drive and differential to meet acceleration, gradeability and maximum speed together; check motor operating on efficiency map for the drive cycle.
- Hybrids: choose gear ratio between engine and wheels so that engine rpm at cruise lies near the optimum BSFC

region.

6. Selection of Energy Storage Technology

Energy storage options: **Li-ion / NiMH batteries**, **ultracapacitors**, **flywheels**, **fuel cell** (source), or a **combination** (hybrid storage). Selection depends on required **energy** (range) and **power** (acceleration / regen).

Device	Specific energy	Specific power	Life	Suitable for
Li-ion	100-250 Wh/kg	300-1500 W/kg	1000-3000 cyc	BEV, PHEV, HEV
NiMH	60-120 Wh/kg	200-1000 W/kg	500-1500 cyc	HEV
Ultracapacitor	3-10 Wh/kg	5000+ W/kg	>500 000 cyc	Peak power, regen
Flywheel	10-50 Wh/kg	~5000 W/kg	Very long	Buses, special
Fuel cell + H_2	High (system)	Low-medium	5000 h+	FCEV, range

Sizing of the battery pack

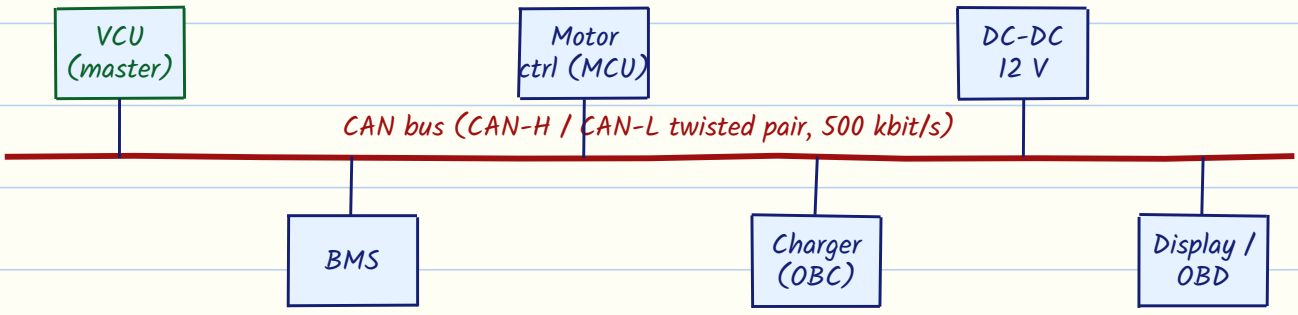
- **Energy**: E_{pack} = (range × Wh/km) / usable DoD. Example: 150 km × 150 Wh/km = 22.5 kWh; with 80 % usable window -> **28 kWh** pack.

- **Power**: $P_{batt,max} \geq P_{motor,peak} / (\eta_m \eta_c)$ and also ≥ regen power limit. For HEVs power, not energy, sizes the battery (1-2 kWh with high C-rate); for EVs energy dominates.
- **Series-parallel arrangement**: $N_s = V_{pack}/V_{cell}$; capacity Ah = E_{pack}/V_{pack}. Example: 360 V / 3.6 V = 100 cells in series; 28 kWh / 360 V = 78 Ah -> 78 Ah cells (or e.g. 2 parallel strings of 39 Ah). Mass = 28 000 / 150 Wh/kg ≈ **187 kg**.
- Add BMS, cooling, contactors, fuse; check C-rate (P / E) and cycle life; consider **hybrid storage** (battery + ultracap) to reduce peak stress and extend battery life.

7. Communications and Supporting Subsystems

Communications

In-vehicle communication network



- Many ECUs must share data: **VCU** (vehicle control unit) gets pedal, brake, gear and SOC signals and sends torque commands to the motor controller and engine ECU; BMS

reports voltage, SOC, temperature; charger and DC-DC report status.

- Protocols: **CAN (ISO 11898)** most common, **LIN** (low-cost body control), **FlexRay** (fast, safety-critical), **Ethernet** for infotainment, **PLC / ISO 15118** and CHAdeMO / CCS for charger communication.

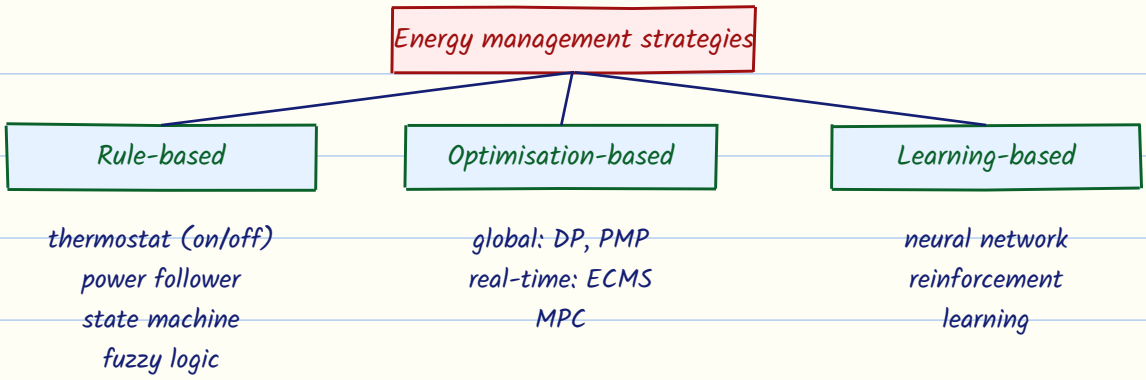
Supporting subsystems

- **Auxiliary power supply**: DC-DC converter (high voltage -> 12 V) for lamps, ECUs, wipers; 12 V battery retained for safety.
- **Thermal management**: separate cooling loops (liquid) for battery, motor and inverter; PTC / heat-pump for cabin heating; battery heater for cold climates.
- **Brake system**: **regenerative braking blending** with hydraulic brakes; electric vacuum pump for brake booster; electric power steering (EPS) and electric A/C compressor.
- **Safety**: high-voltage interlock loop (HVIL), isolation monitoring, main contactors with pre-charge, crash disconnect, emergency service plug, fuse; on-board charger and charging port.

8. Energy Management Strategies (EMS) in Hybrid and Electric Vehicles

The **EMS** (supervisory controller) decides at every instant the power split between ICE, motor and battery to minimise fuel (or energy) consumption and emissions while keeping the battery SOC in the safe window and satisfying driver demand. In pure EVs the EMS manages battery / ultracap / fuel-cell sharing and regeneration.

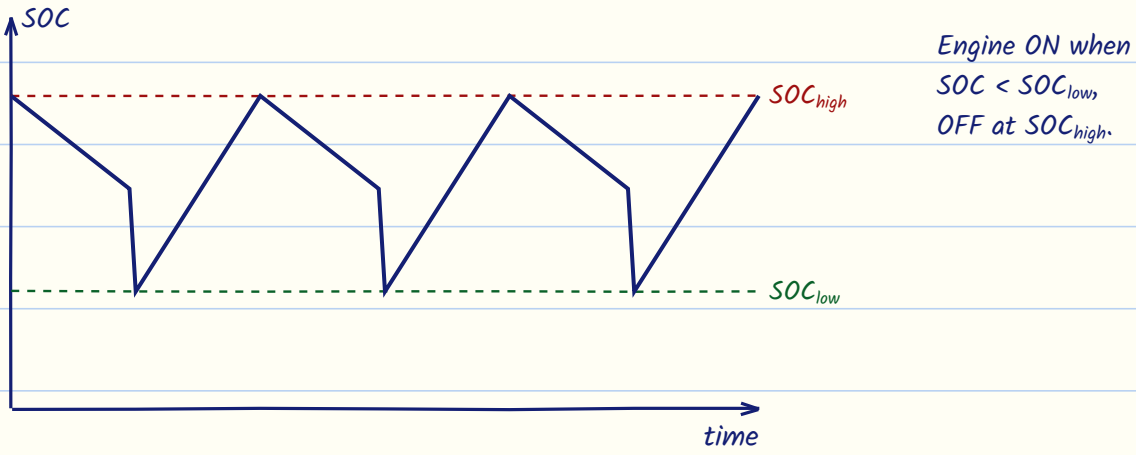
(a) Classification



- **Rule-based (heuristic)**: predefined rules from engineering experience; very simple, real-time, but not optimal.
  - **Thermostat (on-off) control** - engine switched on when SOC < SOC_{low}, run at best-efficiency point until SOC_{high}, then off. Used in series hybrids.
  - **Power follower** - engine follows demand; battery supplies the remaining; **modified** versions keep engine near

optimum line. **State machine / fuzzy logic** handle nonlinearity.

- **Optimisation-based**: minimise a cost function $J = \int (\dot{m}_f + \lambda(SOC)\cdot\dots) \, dt$.
  - **Global**: dynamic programming (DP), Pontryagin's minimum principle (PMP) - optimal but need the whole drive cycle in advance (offline, benchmark).
  - **Real-time**: **ECMS** (equivalent consumption minimisation strategy), model predictive control (MPC), adaptive ECMS.
- **Learning-based / predictive**: neural networks, reinforcement learning, GPS / traffic-based predictive control (preview information).



$$\text{ECMS: } \dot{m}_{eq}(t) = \dot{m}_f(t) + s(t)\, P_{batt}(t) / Q_{LHV}$$

$s(t)$ = equivalence factor converting electric power into equivalent fuel (Q_{LHV} = lower heating value of fuel, 42-44

MJ/kg). At each instant the split minimising $\dot{m}_{eq}$ is chosen.

(b) Comparison

Strategy	Optimality	Computation	Needs trip info	Real-time use
Rule-based	Low-medium	Very low	No	Yes - production cars
DP / PMP	Global optimum	Very high	Yes (full cycle)	No (benchmark)
ECMS	Near optimal	Low-medium	No / partial	Yes
MPC	Near optimal	Medium-high	Preview	Yes
Learning	Good, adaptive	Training heavy	Optional	Yes (after training)

(c) Implementation

- Run on the VCU (microcontroller) as periodic task (10-100 ms). **Inputs**: driver pedal demand, vehicle speed, SOC, battery temperature, engine speed and temperature, road / trip data. **Outputs**: torque commands to engine and motor, mode (EV / hybrid / regen), battery charge limit.
- Rules or look-up tables (offline-optimised maps) are

stored in memory; **SOC feedback** keeps charge sustaining (adjust s(t) or thresholds).

- Development path: model-in-the-loop (MIL, Simulink) -> software-in-the-loop (SIL) -> **hardware-in-the-loop (HIL)** -> vehicle testing on chassis dynamometer; auto-code generation (Embedded Coder / dSPACE), then calibration.
- Add constraints: drivability (no torque jerk during engine start), emissions / catalyst temperature, battery life (limit C-rate and DoD), safety fallback (limp-home mode).

Important questions: (1) Matching of ICE and electric machine. (2) Sizing of propulsion motor with example. (3) Sizing of battery and power electronics. (4) Communication and supporting subsystems. (5) Classify and compare EMS strategies (ECMS, rule-based, DP).